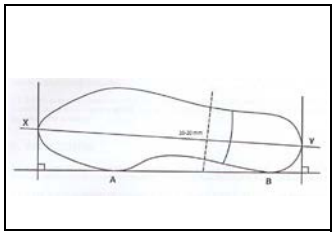
STANDARD OPERATING PROCEDURE OF BATA BELT VF-D03

Lab /SOP/ 21



Bata Belt Flex machine
Belt of cotton canvas
Cutting devise, such as sharp knife or scalpel
T-square, with an integral angle of 90 ±1 °
Caliper to measure the crack



Make a center line from the toe to heel on
the outer surface of the outsole
Place the points A and B, on the inside
edge of the outsole against the longer edge of
the T square and the toe against the shorter edge.
Make a mark on the toe at the point that is in
contact with the shorter edge of the T square.



Repeat the procedure in the last sections for the
heel of the specimen.
Mark a line between the point at the toe
and the point at the heel.
Cut off the heel and part of the waist
portion so as to leave 10 to 20mm length
of the waist on the forepart.



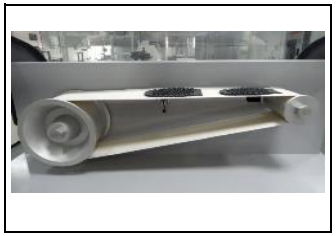
Place the specimens to be tested onto the belt
Mark the perimeter of each specimen on the belt.
Attach the specimen to the belt
Condition or store the prepared belt in standard
controlled environment of 23±2℃ and 55±10%
relative humidity for more than 16 hours to
allow the adhesive to set before your test.



Select a proper smaller roller as per sole units.
90mm in diameter suitable for typical sole units.
60mm in diameter suitable for very flexible sole units.
120mm in diameter suitable for very rigid sole units.
Fit the prepared belt over the larger roller, and the
smaller roller.



Set the number of flexing cycles according to
the test requirement
Press the START button and when the set value
has been reached, the machine will stop automatically.



Inspect the specimens in every 5,000
cycles, until the required number of cycles
is reached.
Slowly move the belt round by hand and inspect
the surface of each specimen for signs of
cracking as it passes over the smaller roller.
If any cracks are found, then record their
position, and their length and depth to the nearest 1mm.



Appearance of cracks, record their
position, and their length and depth to the
nearest 1mm.
Number of flexing cycles.

Prepared by	Approved by	REVISION		DATE	